Tutorial 10 Solution

Problem 1

Consider the maximization problem

$$\max_{(x,y)\in\mathbb{R}^2} -(x_1-4)^2-(x_2-4)^2 \qquad \text{subject to} \quad x_1+x_2\leq 4 \quad \text{and} \quad x_1+3x_2\leq 9.$$

(a) Are the Lagrange conditions necessary? That is, is it true that every solution of the maximization problem, if exists, must satisfy the Lagrange conditions?

(b) Write down the Lagrange conditions for this problem.

(c) Find all points that satisfy the Lagrange conditions.

Solution:

(a) The Lagrange conditions are necessary because all constraint functions are affine.

(b) The Lagrange conditions are

$$\begin{aligned}
&-2(x_1-4)-\lambda_1-\lambda_2=0\\
&-2(x_2-4)-\lambda_1-3\lambda_2=0\\
&x_1+x_2\leq 4,\lambda_1\geq 0, \text{ and } \lambda_1(x_1+x_2-4)=0\\
&x_1+3x_2\leq 9,\lambda_2\geq 0, \text{ and } \lambda_2(x_1+3x_2-9)=0
\end{aligned}$$

(c) We start from the two complementary slackness conditions, $\lambda_1(x_1+x_2-4)=0$ and $\lambda_2(x_1+3x_2-9)=0$, by discussing the signs of λ_1 and λ_2. There are four cases:

- $\lambda_1=\lambda_2=0$. In this case, the first two equations of the Lagrange conditions above indicate that $x_1=x_2=4$, but this violates the constraint $x_1+x_2\leq 4$, and therefore can't be the case.
- $\lambda_1>0$ and $\lambda_2=0$. In this case, the first two equations of the Lagrange conditions above indicate that
$$x_1=x_2=4-\lambda_1/2.$$
Because $\lambda_1>0$, complementary slackness implies that $x_1+x_2=4$. Plugging $x_1=x_2$ in, we get $x_1=x_2=2$, which also satisfies $x_1+3x_2\leq 9$. Thus, $(x_1^*,x_2^*,\lambda_1^*,\lambda_2^*)=(2,2,4,0)$ satisfies the Lagrange conditions.
- $\lambda_1=0$ and $\lambda_2>0$. In this case, the first line of the Lagrange conditions can be written as $\lambda_2=-2(x_1-4)$; plug into the second line, we get
$$-2(x_2-4)+6(x_1-4)=0 \quad \Leftrightarrow \quad 3x_1-x_2=8.$$
Now we have two equations and two unknowns: $3x_1-x_2=8$, $x_1+3x_2=9$; the solution is $x_1=3.3$ and $x_2=1.9$. However, $x_1+x_2=5.2>4$, which can't be the case.

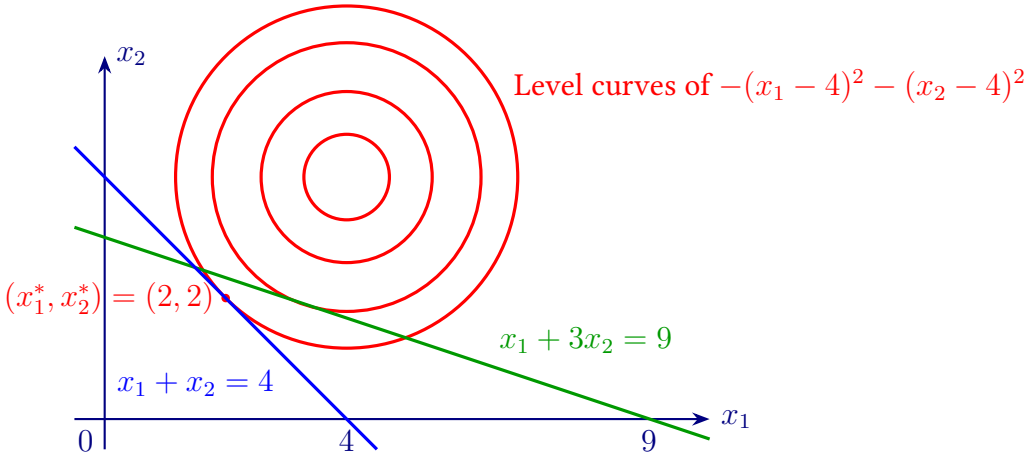
- $\lambda_1 > 0$ and $\lambda_2 > 0$. Complementary slackness imply that we have two equations and two unknowns: $x_1 + x_2 = 4$, $x_1 + 3x_2 = 9$; the solution is $x_1 = 1.5$ and $x_2 = 2.5$. Then the first two equations of the Lagrange conditions become

$$\begin{aligned} 5 - \lambda_1 - \lambda_2 &= 0 \\ 3 - \lambda_1 - 3\lambda_2 &= 0 \end{aligned}$$

  implying that $\lambda_1 = 6$ and $\lambda_2 = -1$, which violates the condition $\lambda_2 \geq 0$.

Thus, the unique point that satisfies the Lagrange conditions is $(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*) = (2, 2, 4, 0)$.

Note. We can solve this problem graphically:



Indeed, the unique solution is $(x_1^*, x_2^*) = (2, 2)$, confirming that every solution must satisfy the Lagrange conditions.

Problem 2

Consider the maximization problem

$$\max_{(x,y)\in\mathbb{R}^2} x - y \qquad \text{subject to} \quad x^2 + y^2 = 8$$

(a) Does a solution to this maximization problem exist? Explain.

(b) Find all solutions to this maximization problem.

Solution:

(a) Since the objective function is linear, it is continuous. Moreover, the constraint set C is compact because it is the circle centered at the origin with radius $2\sqrt{2}$. By the extreme value theorem, a solution exists.

(b) Define the constraint function $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) = x^2+y^2-8$. Note that $\nabla g(x, y) \neq (0, 0)$ for all $(x, y) \in C = \{(x, y) \in \mathbb{R}^2 : \; x^2 + y^2 = 8\}$ as $\nabla g(x, y) = (2x, 2y) = (0, 0)$ only if $(x, y) = (0, 0) \notin C$. Thus, $\{\nabla g(x, y)\}$ is linearly independent for all $(x, y) \in C$. Therefore, the Lagrange conditions are necessary; i.e., every solution to the maximization problem must satisfy the Lagrange conditions.

The Lagrange conditions are

$$\begin{aligned} 1 - 2\lambda x &= 0 \\ -1 - 2\lambda y &= 0 \\ x^2 + y^2 - 8 &= 0 \end{aligned}$$

From the first equation, $\lambda \neq 0$. Therefore, the first two equation together imply that $y = -x$. The third equation then implies $2x^2 = 8$ and $x = \pm 2$. Consequently, either $x = 2$ and $y = -2$, or $x = -2$ and $y = 2$. Plugging $x = 2$ and $x = -2$ into the first equation, we get $\lambda = 1/4$ and $\lambda = -1/4$, respectively. Thus, there are two points satisfying the Lagrange conditions:

$$\begin{aligned} (x_1^*, y_1^*, \lambda_1^*) &= \left(2, -2, \frac{1}{4}\right) \\ (x_2^*, y_2^*, \lambda_2^*) &= \left(-2, 2, -\frac{1}{4}\right) \end{aligned}$$

Evaluate the function at $(x_1^*, y_1^*) = (2, -2)$ and $(x_2^*, y_2^*) = (-2, 2)$:

$$\begin{aligned} f(x_1^*, y_1^*) &= f(2, -2) = 4 \\ f(x_2^*, y_2^*) &= f(-2, 2) = -4 \end{aligned}$$

Because the Lagrange conditions are necessary and a solution exists by part (a), the points satisfy the Lagrange conditions and yield the highest value of the objective function is the solutions. Thus, $(x_1^*, y_1^*) = (2, -2)$ is the unique solution.

Problem 3

(a) For $x > 0, y > 0, z > 0$, write down the Lagrange conditions the constrained maximization problem

$$\max_{(x,y,z)\in\mathbb{R}^3_+} 2\ln x + 5\ln y + \ln z \qquad \text{s.t.} \quad x^2 + y^2 + z = 32,$$

and find the unique point satisfying the Lagrange conditions.

(b) *Assuming that the point you found is indeed the unique solution to the maximization problem above*, use your answer to part (a) to provide an approximation of the optimal value of the maximization problem

$$\max_{(x,y,z)\in\mathbb{R}^3} 2\ln x + 5\ln y + \ln z \qquad \text{s.t.} \quad x^2 + y^2 + z = 32.1$$

Do *not* solve this maximization problem. Briefly explain your reasoning.

(c) (**Extra Practice Problem**) For $x \geq 2, y \geq 2, z = 3$, solve (if there is a solution) the constrained minimization problem

$$\min_{x\geq 2, y\geq 2, z>0} 2\ln x + 5\ln y + \ln z \qquad \text{s.t.} \quad x^2 + y^2 + z = 32; \quad z = 3$$

If there is no solution, explain why (you can assume that every solution of the problem must satisfy the Lagrange conditions).

Solution:

(a) Letting $g(x,y,z) = x^2 + y^2 + z$ then $\nabla g(x,y,z) = (2x, 2y, 1) \neq (0,0,0)$. Therefore, $\{\nabla g(x,y,z)\}$ is linearly independent for every (x,y,z) in the constraint set. Thus, the Lagrange conditions are necessary. The Lagrange conditions are

$$\begin{aligned}
&\frac{2}{x} - 2\lambda x = 0 \quad \Leftrightarrow \quad x^2 = \frac{1}{\lambda} \\
&\frac{5}{y} - 2\lambda y = 0 \quad \Leftrightarrow \quad y^2 = \frac{5}{2\lambda} \\
&\frac{1}{z} - \lambda = 0 \quad \Leftrightarrow \quad z = \frac{1}{\lambda} \\
&x^2 + y^2 + z - 32 = 0
\end{aligned}$$

Plugging the first three equations into the last gives

$$1 + \frac{5}{2} + 1 - 32\lambda = 0 \quad \Leftrightarrow \quad \lambda = \frac{9}{64};$$

plugging this back to the first three equations, we get that the unique point satisfying the Lagrange condition is

$$(x^*, y^*, z^*, \lambda^*) = \left(\frac{8}{3}, \frac{8\sqrt{5}}{3\sqrt{2}}, \frac{64}{9}, \frac{9}{64}\right).$$

(b) Let $V(32)$ be the optimal value of the problem in part (a), and let $V(32.1)$ be the optimal value of the problem in this part. Using the multiplier λ^* corresponding to the solution from part (a), and recall that λ^* can be interpreted as the per unit increase in objective function value from the constraint for a sufficiently small amount:

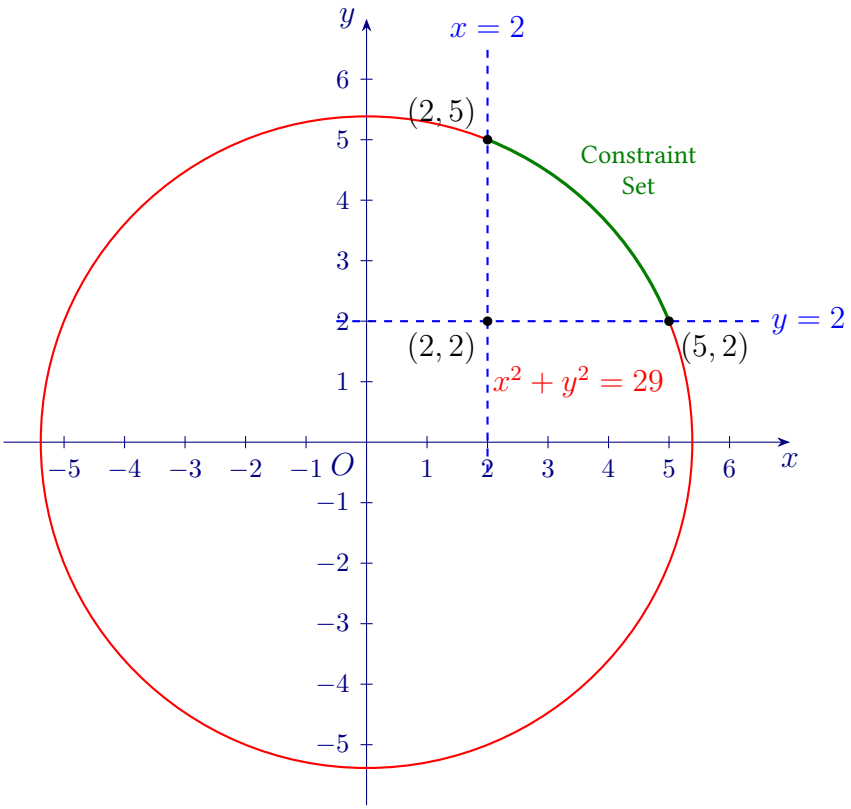
$$V(32.1) \approx V(32) + 0.1\lambda^* = 2(\ln 8 - \ln 3) + 5(\ln 8 + \ln\sqrt{5} - \ln 3 - \ln\sqrt{2}) + \ln 64 - \ln 9 + \frac{9}{640}$$

(The exact calculation doesn't really matter; what matters is the first part.)

(c) The problem is equivalent to

$$\max_{(x,y)\in\mathbb{R}^2} -2\ln x - 5\ln y - \ln 3 \qquad \text{s.t.} \quad x^2 + y^2 = 29; \quad 2 - x \leq 0; \quad 2 - y \leq 0$$

The constraint set is visualized below; we see that it is compact. Because the objective function is continuous, a solution exists.



The Lagrangian conditions are (for a reference, see the last slide in lecture 10)

$$\begin{aligned} -\frac{2}{x} - 2\lambda_1 x + \lambda_2 &= 0 \\ -\frac{5}{y} - 2\lambda_1 y + \lambda_3 &= 0 \\ x^2 + y^2 - 29 &= 0 \\ \lambda_2(2 - x) &= 0 \\ \lambda_3(2 - y) &= 0 \end{aligned}$$

Again, we tackle the problem by starting from the two complementary slackness conditions, namely the last two equations above.

If $\lambda_2 > 0$ and $\lambda_3 > 0$, the complementary slackness conditions imply $x = y = 2$, which violates the equality constraint $x^2 + y^2 = 29$, a contradiction.

If $\lambda_2 = \lambda_3 = 0$, the first two equations above imply that $x^2 = -\frac{1}{\lambda_1}$ and $y^2 = -\frac{5}{2\lambda_1}$. Then the equality constraint implies that $1 + \frac{5}{2} = -29\lambda_1$, yielding $\lambda_1 = -\frac{7}{58}$, which in turn yields $x = \sqrt{\frac{58}{7}}$ and $y = \sqrt{\frac{145}{7}}$. Thus, $(x^*, y^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(\sqrt{\frac{58}{7}}, \sqrt{\frac{145}{7}}, -\frac{7}{58}, 0, 0\right)$ satisfies the Lagrange conditions.

Now suppose $\lambda_2 = 0$ and $\lambda_3 > 0$. The last equality then implies $y = 2$; from the equality constraint $x = 5$. Plugging these into the first two equalities, we get $\lambda_1 = -\frac{1}{25}$ and $\lambda_3 = \frac{117}{50}$. Thus, $(x^*, y^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(5, 2, -\frac{1}{25}, 0, \frac{117}{50}\right)$ satisfies the Lagrange conditions.

Finally, suppose $\lambda_3 = 0$ and $\lambda_2 > 0$. The penultimate equality implies that $x = 2$ and hence from the equality constraint $y = 5$. Plugging these into the first two equalities,

we get Also $\lambda_1 = -\frac{1}{10}$ and $\lambda_2 = \frac{3}{5}$. Thus, $(x^*, y^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(2, 5, -\frac{1}{10}, \frac{3}{5}, 0\right)$ satisfies the Lagrange conditions.

The value of the objective function at the three candidate solutions are

$$f\left(\sqrt{\frac{58}{7}}, \sqrt{\frac{145}{7}}\right) = -2\ln\sqrt{\frac{58}{7}} - 5\ln\sqrt{\frac{145}{7}} - \ln 3$$
$$f(5,2) = -2\ln 5 - 5\ln 2 - \ln 3$$
$$f(2,5) = -2\ln 2 - 5\ln 5 - \ln 3$$

Since a solution exists, and every solution must satisfy the Lagrange conditions, the points which the objective function obtains the highest value are solutions. Comparing the values, we reach the conclusion that the unique solution is $(x^*, y^*) = (5, 2)$.

Problem 4 (Extra Practice Problem)

Consider the maximization problem

$$\max_{(x,y)\in\mathbb{R}^2} 2 - (x-1)^2 - e^{y^2} \qquad \text{subject to} \quad x^2 + y^2 \leq a.$$

Write down the Lagrange conditions, then find the only point that satisfies the Lagrange conditions.
(*Hint.* Distinguishing between the cases $a \in (0,1)$ and $a \geq 1$.)

Solution: The Lagrange conditions are

$$\begin{aligned} -2(x-1) - 2\lambda x &= 0 \\ -2ye^{y^2} - 2\lambda y &= 0 \\ x^2 + y^2 &\leq a \\ \lambda &\geq 0 \\ \lambda(x^2 + y^2 - a) &= 0 \end{aligned}$$

The first two equations can be simplified to

$$(x-1) + \lambda x = 0, \tag{1}$$
$$2y(e^{y^2} + \lambda) = 0. \tag{2}$$

We analyze the solution by considering two cases based on the complementary slackness condition, namely the last equation in the Lagrange conditions.

Case 1: The constraint is not binding ($\lambda = 0$). If $\lambda = 0$, the complementary slackness condition is satisfied.

- From equation (2): $2y(e^{y^2} + 0) = 0 \implies 2ye^{y^2} = 0$. Since $e^{y^2} > 0$, we must have $y = 0$.
- From equation (1): $(x-1) + 0 \cdot x = 0 \implies x - 1 = 0 \implies x = 1$.

This gives the candidate point $(1, 0)$. The constraint requires

$$1^2 + 0^2 \leq a \implies 1 \leq a$$

So, if $a \geq 1$, the point $(x^*, y^*, \lambda^*) = (1, 0, 0)$ satisfies Lagrange conditions; otherwise it doesn't.

Case 2: The constraint is binding ($\lambda > 0$). If $\lambda > 0$, the complementary slackness condition requires $x^2 + y^2 - a = 0$.

- From equation (2), $2y(e^{y^2} + \lambda) = 0$. Since $\lambda > 0$ and $e^{y^2} > 0$, their sum is positive. Therefore, we must have $y = 0$.
- Substitute $y = 0$ into the binding constraint: $x^2 + 0^2 = a \implies x = \pm\sqrt{a}$.
- We use equation (1) to find the valid value for x:

$$(x - 1) + \lambda x = 0 \implies \lambda = \frac{1 - x}{x}$$

- Since we are in the case where $\lambda > 0$, we require $\frac{1-x}{x} > 0$.
  - Testing $x = \sqrt{a}$: We need $\lambda = \frac{1-\sqrt{a}}{\sqrt{a}} > 0$. Since $\sqrt{a} > 0$ (for $a > 0$), this requires $1 - \sqrt{a} > 0 \implies 1 > a$.
  - Testing $x = -\sqrt{a}$: We need $\frac{1-(-\sqrt{a})}{-\sqrt{a}} > 0$, which is $\frac{1+\sqrt{a}}{-\sqrt{a}} > 0$. This is never true as the numerator is positive and the denominator is negative.

The only valid candidate point in this case is $(x^*, y^*, \lambda^*) = (\sqrt{a}, 0, (1/\sqrt{a} - 1)))$, which satisfies the Lagrange conditions for $0 \leq a < 1$.

To summarize,

- If $a \geq 1$, the unique point satisfying the Lagrange conditions is $(x^*, y^*, \lambda^*) = (1, 0, 0)$ with a maximum value of 1.
- If $0 \leq a < 1$, the unique point satisfying the Lagrange conditions is $(x^*, y^*, \lambda^*) = (\sqrt{a}, 0, (1/\sqrt{a} - 1))$.